Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

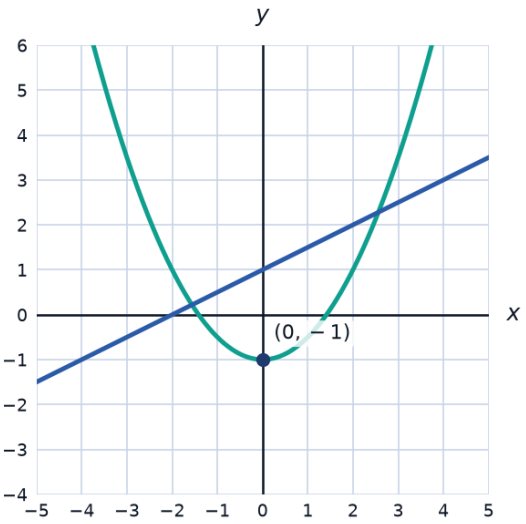
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic arithmetic with complex numbers**

Given $z_1 = 4 + 3i$ and $z_2 = 2 - 5i$, find $z_1 + z_2$ and $z_1 - z_2$.

2 **[Foundational]** • **Target: Basic arithmetic with complex numbers**

Find $z_1 z_2$ for $z_1 = 2 + 3i$ and $z_2 = 4 - i$.



3 **[Foundational]** • **Target: The complex conjugate and division**

Find the conjugate of $z = 7 - 2i$, and compute $z\bar{z}$.

4 **[Foundational]** • **Target: The complex conjugate and division**

Simplify $\frac{3+i}{2-3i}$ by multiplying by the conjugate of the denominator.

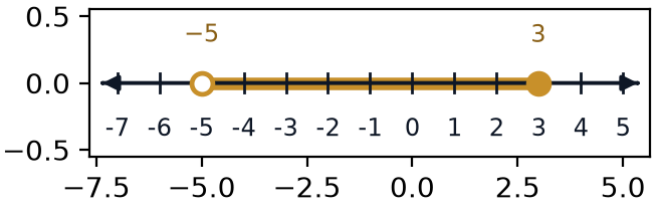
5 **[Foundational]** • **Target: Modulus and the Argand diagram**

Find the modulus $|z|$ of $z = 5 - 12i$.

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6 [Foundational] • Target: Modulus and the Argand diagram

The graph shows several complex numbers plotted on the Argand diagram (real part horizontal, imaginary part vertical).



a Read off the complex number plotted at $(4, 1)$ in Cartesian form.

b Find the modulus of the complex number plotted at $(-2, -3)$.

7 [Foundational] • Target: Solving equations with complex roots

Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving equations with complex roots

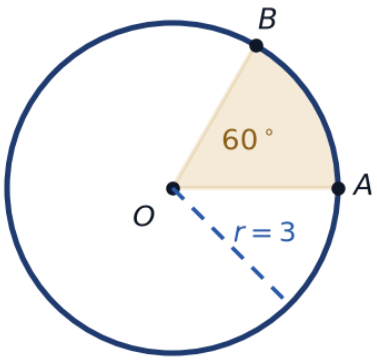
One root of a quadratic equation with real coefficients is $3 - 4i$. State the other root, and find the quadratic equation.

9 [Intermediate] • Target: Solving equations with complex roots

Given that $z = 1 + 2i$ is a root of $z^3 - 4z^2 + 9z - 10 = 0$ (with real coefficients), find the other two roots.

10 [Intermediate] • Target: Converting to polar (modulus-argument) form

Write $z = \sqrt{3} + i$ in polar form $r(\cos\theta + i\sin\theta)$, giving the argument in radians.



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11 [Intermediate] • Target: Converting to polar (modulus-argument) form

Convert $z = 6\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ to Cartesian form.

12 [Intermediate] • Target: Multiplying and dividing in polar form

Given $z_1 = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $z_2 = 2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

13 [Intermediate] • Target: Multiplying and dividing in polar form

Given $z_1 = 5\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)$ and $z_2 = \cos\frac{\pi}{5} + i\sin\frac{\pi}{5}$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

14 [Intermediate] • Target: Multiplying and dividing in polar form

Find the modulus and argument of $z = \frac{(1+i)^3}{1-i}$, using De Moivre's theorem for the numerator and polar division.

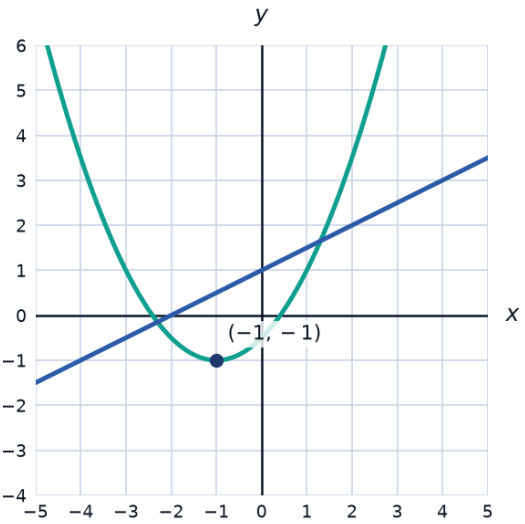
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Use De Moivre's theorem to find $\left[3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^4$, giving your answer in Cartesian form.

16 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 10.01 Numerical Methods

Use De Moivre's theorem to find $(1 - i)^5$, giving your answer in Cartesian form.



17 [Challenging] • Target: Finding roots using De Moivre's theorem • Cross-Lesson: 11.01 Differential Equations

Find the three cube roots of $z = 27\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$, giving arguments in $[0, 2\pi)$.

18 [Challenging] • Target: Finding roots using De Moivre's theorem • Cross-Lesson: 12.01 Poisson Distribution

Find all four fourth roots of -16 (i.e. solve $z^4 = -16$), giving arguments in $[0, 2\pi)$.

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19 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 13.01 Continuous Random Variables

This question has three parts. An AC circuit has impedance $Z = R + iX_L$, where $R = 8\,\Omega$ (resistance) and $X_L = 6\,\Omega$ (inductive reactance). (a) Write Z in Cartesian form, find $|Z|$, and find $\arg Z$ to the nearest degree. (b) The current is $I = \frac{V}{Z}$, where $V = 100$ volts (real). Find I in the form $a + bi$, by rationalizing the denominator. (c) Find $|I|$, to two decimal places.

20 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 14.01 Sampling & Estimation

The complex number z is given by $z = \frac{4+3\mathrm{i}}{2-\mathrm{i}}$. Express z in the Cartesian form $x + \mathrm{i}y$, where x and y are real numbers.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Cartesian Form & Complex Arithmetic

Given $z = 3 - 4i$, compute $z \cdot z^*$ where z^* is the complex conjugate of z.

a. 25

b. 7

c. -7

d. $25 - 24i$

22 [Foundational] • Target: Cartesian Form & Complex Arithmetic

Express $\frac{5+2i}{2-i}$ in the form $a + bi$.

a. $\frac{8}{5} + \frac{9}{5}i$

b. $\frac{12}{5} + \frac{9}{5}i$

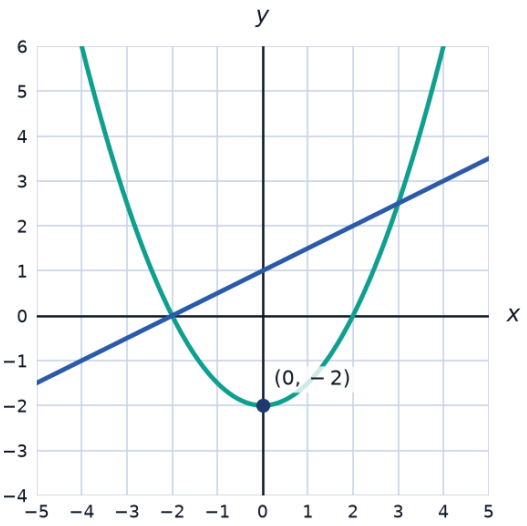
c. $8 + 9i$

d. $\frac{8}{3} + \frac{9}{3}i$

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23 [Foundational] • Target: Cartesian Form & Complex Arithmetic

Find the two square roots of $-5 + 12i$.



a. $\pm(2 + 3i)$

b. $\pm(3 + 2i)$

c. $\pm(2 - 3i)$

d. $\pm(3 - 2i)$

24 [Foundational] • Target: Modulus-Argument Form

Find the modulus and principal argument of $z = -1 + i\sqrt{3}$.

a. $r = 2, \theta = \frac{2\pi}{3}$

b. $r = 2, \theta = \frac{\pi}{3}$

c. $r = 4, \theta = \frac{2\pi}{3}$

d. $r = 2, \theta = -\frac{2\pi}{3}$

25 [Foundational] • Target: Modulus-Argument Form

Given $z_1 = 3(\cos(\pi/4) + i\sin(\pi/4))$ and $z_2 = 2(\cos(\pi/6) + i\sin(\pi/6))$, find $\arg(z_1 z_2)$.

a. $\frac{5\pi}{12}$

b. $\frac{\pi}{12}$

c. $\frac{\pi}{24}$

d. $\frac{7\pi}{12}$

26 [Foundational] • Target: Modulus-Argument Form

Express $z = 4e^{i\pi/3}$ in Cartesian form $x + iy$.

a. $2 + 2i\sqrt{3}$

b. $2\sqrt{3} + 2i$

c. $2 - 2i\sqrt{3}$

d. $4 + 4i\sqrt{3}$

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27 [Foundational] • Target: Loci in the Argand Plane

Describe the locus of points represented by $|z - 3 - 4i| = 5$ in the complex plane.

a. A circle with center $(3, 4)$ and radius 5

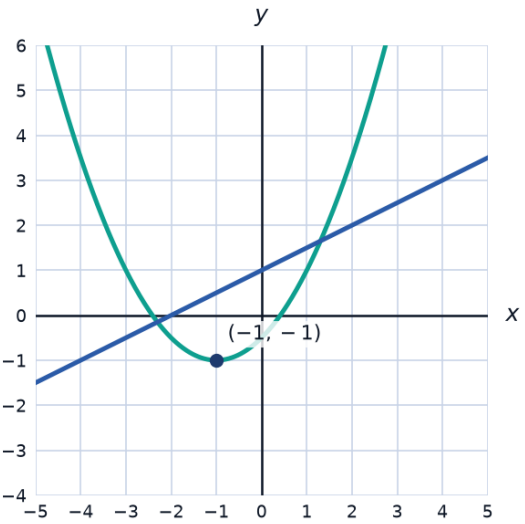
b. A circle with center $(-3, -4)$ and radius 5

c. A circle with center $(3, 4)$ and radius 25

d. A straight line perpendicular to $3 + 4i$

28 [Foundational] • Target: Loci in the Argand Plane

Describe the locus of points satisfying $|z - 2| = |z - 6i|$.



a. Perpendicular bisector of the segment joining $(2, 0)$ and $(0, 6)$

b. Circle passing through $(2, 0)$ and $(0, 6)$

c. Ellipse with foci at $(2, 0)$ and $(0, 6)$

d. Line segment joining $(2, 0)$ and $(0, 6)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Loci in the Argand Plane

Describe the geometric locus defined by $\arg(z-2)=\frac{\pi}{4}$.

a. A half-line starting from $(2,0)$ making an angle of $\pi/4$ with the positive real axis

b. A full line through $(2,0)$ with gradient 1

c. A circle of radius 2

d. A ray towards the origin

30 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Given that $z=1+2i$ is a root of $z^3-3z^2+7z-5=0$, what are the other two roots?

a. $z=1-2i$ and $z=1$

b. $z=-1-2i$ and $z=1$

c. $z=1-2i$ and $z=5$

d. $z=-1+2i$ and $z=-1$

31 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Find a quadratic equation with real coefficients having root $z=4-3i$.

a. $z^2-8z+25=0$

b. $z^2+8z+25=0$

c. $z^2-8z+7=0$

d. $z^2-6z+25=0$

32 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Find the least value of $|z|$ for points z satisfying $|z-5-12i|\le 3$.

a. 10

b. 13

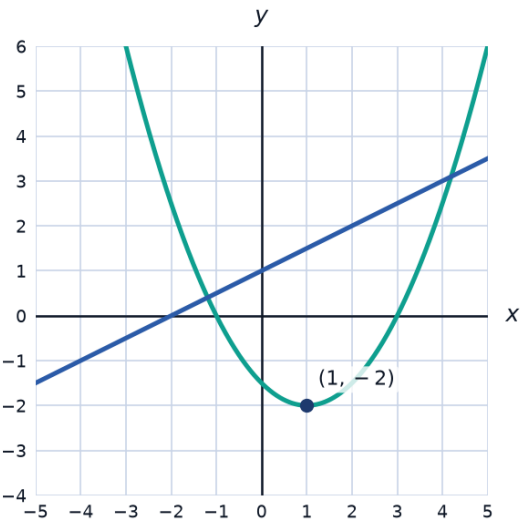
c. 16

d. 7

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33 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Describe the locus $|z-(-11-10i)|=3$ in the Argand diagram. Find its radius.



a. 5

b. 3

c. 6

d. 4

34 [Intermediate] • Target: Basic arithmetic with complex numbers

Describe the locus $|z-(-2+2i)|=3$ in the Argand diagram. Find its radius.

a. 4

b. 3

c. 6

d. 2

35 [Intermediate] • Target: The complex conjugate and division

Describe the locus $|z-(-6-3i)|=5$ in the Argand diagram. Find its radius.

a. 7

b. 8

c. 6

d. 5

36 [Intermediate] • Target: Modulus and the Argand diagram

Describe the locus $|z-(-8-3i)|=9$ in the Argand diagram. Find its radius.

a. 11

b. 9

c. 10

d. 12

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37 [Intermediate] • Target: Solving equations with complex roots

Describe the locus $|z-(1+2i)|=10$ in the Argand diagram. Find its radius.

a. 2

b. 13

c. 10

d. 1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Converting to polar (modulus-argument) form • Cross-Lesson: 12.01 Poisson Distribution

Describe the locus $|z-(3-10i)|=1$ in the Argand diagram. Find its radius.

a. 5

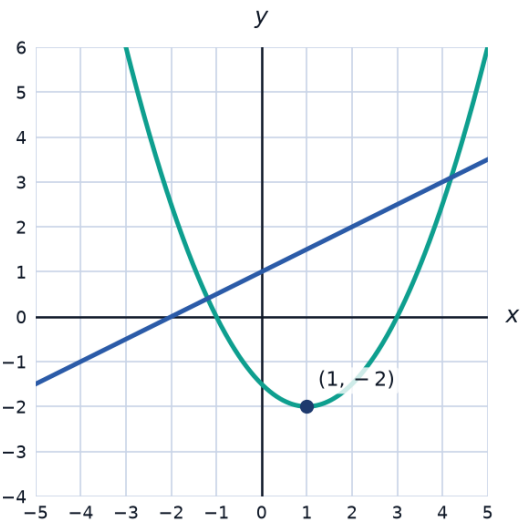
b. 3

c. 1

d. 4

39 [Challenging] • Target: Multiplying and dividing in polar form • Cross-Lesson: 13.01 Continuous Random Variables

Describe the locus $|z-(-1-6i)|=1$ in the Argand diagram. Find its radius.



a. 3

b. 1

c. 4

d. 2

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40 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 14.01 Sampling & Estimation

Describe the locus $|z - (1 + 5i)| = 2$ in the Argand diagram. Find its radius.

a. 2

b. 1

c. 6

d. 5

41 [Challenging] • Target: Finding roots using De Moivre's theorem • Cross-Lesson: 15.01 Hypothesis Testing

Describe the locus $|z - (7 - 6i)| = 9$ in the Argand diagram. Find its radius.

a. 9

b. 11

c. 12

d. 7

42 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 2.01 Exponentials & Logarithms

Describe the locus $|z - (6 + 8i)| = 4$ in the Argand diagram. Find its radius.

a. 6

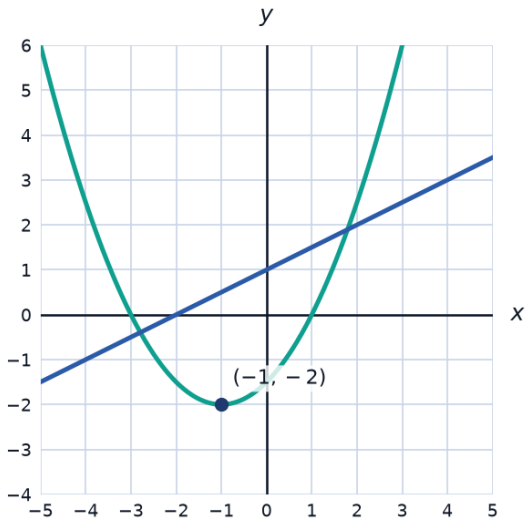
b. 7

c. 8

d. 4

43 [Challenging] • Target: Cartesian Form & Complex Arithmetic • Cross-Lesson: 3.01 Advanced Trigonometry

Describe the locus $|z - (3 - 6i)| = 8$ in the Argand diagram. Find its radius.



a. 8

b. 11

c. 3

d. 10

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44 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 4.01 Differentiation

Describe the locus $|z - (6 + 3i)| = 2$ in the Argand diagram. Find its radius.

a. 3

b. 2

c. 5

d. 6

45 [Challenging] • Target: Loci in the Argand Plane • Cross-Lesson: 5.01 Integration

Describe the locus $|z - (2i)| = 1$ in the Argand diagram. Find its radius.

a. 1

b. 4

c. 2

d. 5

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